MODELLING AND ANALYSIS OF MICROSTRUCTURES OF ELECTRONIC MARKETS

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Avanish Gadhikar, Kushal Gupta

Roll No. 220123075, 220123082

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CERTIFICATE

This is to certify that the work contained in this project report entitled "Modelling and Analysis of Microstructures of Electronic Markets" submitted by Avanish Gadhikar, Kushal Gupta (Roll Nos.: 220123075, 220123082) to the Department of Mathematics, Indian Institute of Technology Guwahati towards partial requirement of Bachelor of Technology in Mathematics and Computing has been carried out by them under my supervision.

It is also certified that, along with literature survey, computational implementations and simulation studies have been carried out by the students under the project.

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(Prof. Siddhartha Pratim Chakrabarty)
Project Supervisor

ABSTRACT

The primary aim of this project is to understand the functioning of electronic exchanges, the driving principles behind the dynamics of such markets, how such dynamics affect the underlying asset prices and and the simulation of such exchanges through models described in literature. The project starts by discussing the need for electronic exchanges and their subsequent advent. Thereafter, the discussion continues on the dynamics of electronic exchanges, primarily the dynamics behind their Order Books. In the first part, we provide the dynamics of the Order Book from the perspective of microstructures of markets. The second part of this project discusses simulation of Limit Order Books and some of their important features through the use of zero intelligence models such as Poisson processes and Hawkes processes, followed by deep learning approaches for the same using GANs and CNN-LSTMs.

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Chapter 1

A Primer on Limit Order Books and Electronic Exchanges

Prior to the advent of electronic markets, the public exchanges were predominantly "quote-driven markets" or "open-outcry markets".

For the quote-driven markets, dealers (or market makers) usually bought and sold stocks from clients via a broker. They would post quotes indicating their prices (quotes) for buying and selling the stock. This system resulted in dealers having to hold the asset, and having access to capital as well as the market. However, this system suffered from the shortcomings of being opaque and inefficient, in addition to having additional transaction costs. NYSE, NASDAQ, LSE among others have all utilised this system or its variation during the last two decades of the 20th century. This system can still be seen utilised by FX market dealers who buy low and sell high, making a profit across the bid-ask spread.

In contrast, open outcry markets usually involve buyers and sellers meet-

ing and advertising their prices competitively in a shared space called a "pit". While this system cuts out the need for dealers and the risk of holding the asset, it introduces new challenges implicit in the form of "skilled traders". These exchanges were often viewed as being inefficient and open to dispute, while also not being transparent enough in the sharing of market data. Exchanges such as the CBOT, CME, LIFFE initially operated an open outcry system before transitioning to an electronic system.

1.1 The Shift to Order Driven Markets

The two preceding systems (evidently) had inherent design flaws and were criticised for low transparency, latency, and heavy reliance on middle-men. This prompted a shift in general towards Order Driven Markets. The Toronto Stock Exchange was the first to introduce a Computer Assisted Trading System (CATS) with order book features in 1977. The Paris Bourse is cited as having the first fully electronic Limit Order Book (LOB) based equity trading system (1986) [1].

A LOB is a setup that the vast majority of organized electronic markets use to store (in their central computer) the list of all the interests of market participants. It contains all orders with characteristics such as: buy/sell side, price, quantity, timestamp, and other market-dependent information. In other words, the LOB contains, at any given point in time, the list of all the transactions that one could possibly perform in the market [1].

A market in which buyers and sellers meet via an LOB is called an order driven market. These are electronic platforms where buyers and sellers can place orders, matched subject to some priority rules. These markets eliminate the need for middlemen, permit clients to act as traders themselves, and

allow viewing of the entire market. The electronic nature enhanced the ease of trading, reduced latency significantly, and thus permitted High Frequency and Algorithmic Trading. Most markets and exchanges now operate on a fully electronic or at least hybrid version of an LOB [1, 3].

1.2 Order Book Dynamics

In order to complete trades, traders submit orders to the market. Orders can be to buy or sell a specified number of assets, either immediately or at a specified price. In order-driven markets, buy and sell orders are matched as they arrive over time, subject to price and then time priority (FIFO). Accordingly there are three types of orders:

- **Limit Order:** Specifies a price at which one is willing to buy/sell a certain number of shares.
- **Market Order:** Executed immediately at the best available matching quote.
- **Cancellation Order:** Cancels an existing limit order.

Impatient traders rely on Market Orders (MOs), while patient traders rely on Limit Orders (LOs). Buy LOs are placed on the bid side and sell LOs on the ask side. The spread is defined as the gap between best bid and best ask, and the mid-price is the average of both. The matching engine enforces well-defined algorithms to execute trades, usually price-time priority. Market orders may "walk the book" by consuming liquidity at multiple price levels.

The evolution of an order book over time describes market movement. Studying LOBs provides insight into price discovery in financial markets, which behave as complex adaptive systems.

The study of LOBs helps to understand price formation, market impact of new information, design trading algorithms (including High Frequency Trading), and aid regulators in ensuring markets are robust. [5]

1.3 Dark Pools

The origins of Dark Pools date back to 1979 when U.S. regulations allowed securities to be traded off the exchange. The first dark pool was started by Instinet in 1986 called "After Hours Cross". By 2012, 40% of trading volume was in dark pools. As of January 2025, reports estimate that approximately 51.8% of trade volume occurs in Dark Pools. Dark pools now exist in large numbers in the U.S., with more than 60 Alternative Trading Systems (ATSs), with their growth being spurred by the High Frequency Trading firms.

1.3.1 What are Dark Pools?

Dark pools are ATSs that do not provide their best-priced orders for inclusion in consolidated data. They allow anonymous trading away from the lit market but execute at prices no worse than National Best-Bid and Offer (NBBO). [4]

The various types of Dark Pools include:

- Bank/Broker Pools
- Independent/Agency Pools
- Market Maker Pools
- Consortium-Sponsored Pools
- Exchange-Based Pools

Exchanges argue that dark pools divert volume away from lit venues. On the other hand, regulators fear impacts on liquidity provision, trading costs, and welfare distribution between institutional and retail investors.

1.4 Regulations Behind Dark Pools

Dark pool trading remains legal as it allows large trades without disrupting markets and aligns with free market principles. The SEC is the primary regulatory authority, with FINRA assisting in monitoring compliance. Dark pool trading is regulated under MiFID and MiFID II.

- Pre-trade transparency: publishing certain information (prices, volumes, execution quality).
- Post-trade transparency: reporting trade details and consolidated dissemination.
- Surveillance and monitoring systems: detecting suspicious activities.

Dark pools provide liquidity and protect institutional strategies, but regulation ensures transparency, stability, and accountability. Their legality does not imply absence of regulation.

Chapter 2

Understanding the Microstructure of Financial Markets

Market Microstructure is defined as "the study of the process and outcomes of exchanging assets under explicit trading rules" [20]. The field analyzes how specific trading mechanisms and structures affect the price formation process. The microstructure literature provides the foundation for studying the practical aspects of asset exchange, including the activities involved (e.g., trading strategies) and the resulting outcomes (e.g., asset prices, volume, risk transfers). Research in this area exploits the structures provided by various trading mechanisms to model how price-setting rules evolve in different markets. Studies focus on two key aspects of trading: liquidity and price discovery, both of which are the primary drivers for the design of effective algorithmic and high-frequency trading strategies.

Focusing on electronic exchanges driven by **LOBs**, market participants can be broadly classified into three categories:

- **Market Makers (MMs):** Professional traders who profit from facilitating exchange by providing liquidity (posting quotes) and exploiting their skills in trade execution. They are typically passive providers who earn the spread.

- **Informed Traders:** Traders who profit from leveraging private information not yet reflected in market prices by trading assets in anticipation of their appreciation or depreciation.

- **Fundamental (or Noise/Liquidity) Traders:** Those whose trading is driven by exogenous factors such as hedging, consumption needs, or portfolio rebalancing, making them relatively insensitive to short-term price movements.

We now explore how the price dynamics and the composition of the bid-ask spread are affected by the interaction between these participants.

2.1 Market Making and the Cost of Immediacy

Market Makers play a crucial role in markets by providing liquidity to participants, typically by quoting prices to buy (bid) and sell (ask) the assets. We primarily focus on the "passive" market maker, who facilitates trades and profits from the bid-ask spread and execution skills. This requires the ability to quickly adapt to evolving market conditions. The other type is the "active" trader, who anticipates market movements and seeks optimal intervention timing.

In an active, competitive market, the spread is determined by various risk factors. Before delving into models of informational asymmetry, we first

examine the issues faced by liquidity providers in managing their inventory risk.

2.1.1 The Grossman-Miller (1988) Market Making Model

A Market Maker's objective is to provide liquidity by accepting one side of a trade, holding the resulting asset inventory (or a short position) for an uncertain period of time until a matching trade allows them to flatten their position. During this time, the MM is exposed to the risk of the asset's price moving against them, leading to a loss upon liquidation.

The Grossman and Miller (1988) model captures this scenario, explaining how MMs obtain a *liquidity premium* that compensates them exactly for the price risk of holding inventory.

Model Setup and Assumptions

We consider a simplified, three-date setting with **n** being the (finite) number of identical, competitive Market Makers (MMs).

All agents start with initial cash W_0. MMs start with no assets; LT1 holds i units; LT2 is short $-i$ units. The asset's cash value at $t = 3$ is modeled as:

$$S_3 = \mu + \epsilon_2 + \epsilon_3 \tag{2.1}$$

where μ is a constant, and ϵ_2, ϵ_3 are independent random variables following $N(0, \sigma^2)$. Crucially, all participants are *risk-averse*, seeking to maximize the expected utility of their final wealth $\mathbb{E}[U(X_3)]$, with the utility function taken to be the exponential form:

$$U(X) = -\exp(-\gamma X), \quad \gamma > 0 \tag{2.2}$$

where γ is the **risk aversion parameter**.

Backward Solution and Equilibrium Pricing

The model is solved recursively, starting from the final date, as follows

Equilibrium at $t = 2$: At $t = 2$, the shock ϵ_2 is realized, and the MMs, LT1, and LT2 interact. Since ϵ_2 is known and LT1's initial sale is balanced by LT2's purchase, the total net asset supply is $i + (-i) = 0$. Given the zero net supply and competition, the equilibrium price S_2 is the expected future value:

$$S_2 = \mathbb{E}[S_3|\epsilon_2] = \mu + \epsilon_2 \tag{2.3}$$

At this price, the optimal final holding for all agents is $q_2^{j,*} = 0$, as there is no compensation needed to hold the asset beyond $t = 2$.

Equilibrium at $t = 1$: At $t = 1$, MMs and LT1 must determine the price S_1 and their optimal holdings $q_1^{j,*}$. They anticipate the price S_2 at $t = 2$ and the risk exposure from ϵ_2. The optimal asset holding for any agent $j \in \{MM, LT1\}$ is found by maximizing the expected utility, yielding:

$$q_1^{j,*} = \frac{\mathbb{E}[S_2|\epsilon_1] - S_1}{\gamma\sigma^2} = \frac{\mu - S_1}{\gamma\sigma^2} \tag{2.4}$$

The market-clearing condition requires that the total demand match the supply, which is i (the quantity LT1 brings to sell):

$$\sum_j q_1^{j,*} = n \cdot q_1^{MM,*} + q_1^{LT1,*} = i$$

Substituting the optimal holdings:

$$(n+1)\left(\frac{\mu - S_1}{\gamma\sigma^2}\right) = i$$

Solving for the equilibrium trading price S_1:

$$S_1 = \mu - \left(\frac{\gamma\sigma^2}{n+1}\right) i \tag{2.5}$$

Economic Interpretation: The Liquidity Discount

The equation for S_1 is the fundamental insight of the GM model. The deviation of the trading price from the fundamental value μ is the **liquidity discount**:

$$|S_1 - \mu| = \left(\frac{\gamma\sigma^2}{n+1}\right) |i| \tag{2.6}$$

This discount (or premium, if $i < 0$) represents the exact compensation MMs receive for bearing the risk of holding the asset until the matching liquidity (LT2) arrives. It isolates **inventory risk** as the first key component of the bid-ask spread. The discount is directly influenced by:

- **Risk Aversion (γ):** A higher γ increases the required compensation, widening the spread.
- **Volatility (σ^2):** Higher price uncertainty increases the inventory risk and thus widens the spread.
- **Competition (n):** An increase in n reduces the required pre-

mium per MM, narrowing the spread.

This model provides the theoretical basis for the concept of "temporary price impact" and measures of market depth (λ), where the price change is proportional to the size of the trade imbalance.

2.2 Trading on an Informational Advantage

The GM model assumes all information is public. We now turn to models that incorporate **informational asymmetry**, where the spread is necessary to compensate MMs for the risk of trading against a party with superior information, a phenomenon known as **adverse selection**.

2.2.1 The Kyle (1985) Model: Price Discovery

The *Kyle (1985) model* analyzes the optimal trading strategy of a single, monopolistic informed trader and how information is incorporated into prices.

Model Setup and Equilibrium Conditions

In this static model, MMs are **risk-neutral** (eliminating inventory risk). The market consists of:

- **Informed Trader (Insider):** Knows the true final value $v \sim N(\mu, \sigma_v^2)$, and chooses trade $x(v)$.

- **Liquidity Traders (Noise):** Have random, exogenous demand $u \sim N(0, \sigma_u^2)$, providing camouflage.
- **Market Makers:** Observe the total order flow $\omega = x(v) + u$. Competitive MMs set the price $S(\omega)$ to ensure zero expected profits, conditional on observed information: $S(\omega) = \mathbb{E}[v|\omega]$.

The insider maximizes their expected profit $\mathbb{E}[x(v - S)]$ while anticipating the MMs' pricing rule. The unique linear Nash equilibrium is characterized by:

1. The MM's pricing rule: $S(\omega) = \mu + \lambda\omega$
2. The Insider's optimal trade: $x^*(v) = \beta(v - \mu)$

Kyle's Lambda (λ) and Price Impact

The equilibrium value for *Kyle's lambda (λ)* is:

$$\lambda = \frac{\sigma_v}{2\sigma_u} \tag{2.7}$$

Kyle's λ is the measure of the permanent price impact of order flow. It quantifies the degree to which price shifts in response to a trade to reflect the information content.

- **Informational Risk:** λ increases with the volatility of the asset's fundamental value (σ_v), as the potential loss from adverse selection is higher.
- **Camouflage:** λ decreases with the volatility of noise trade (σ_u). Higher noise allows the insider to trade more aggressively (higher β) without causing a large price shift, thus maximizing profit.

- **Price Discovery:** The change in price, $\lambda\omega$, represents the instantaneous movement of the price toward the true value v, demonstrating the mechanism by which private information is integrated into public prices.

2.2.2 The Glosten-Milgrom (1985) Model: The Informational Hedge

The Glosten-Milgrom (GMi) model provides a complementary view, focusing specifically on the MM's quote-setting decision when faced with both informed and uninformed traders.

The Spread as a Conditional Expectation

Using a simplified two-state asset value (V_H or V_L), MMs set the bid (b) and ask (a) prices. Competition dictates that MMs set quotes to maintain zero expected profits. This requires the quote to be the conditional expected value of the asset given the trade direction:

$$\text{Ask Price } a = \mathbb{E}[v \mid \text{Buy Order}] \tag{2.8}$$

$$\text{Bid Price } b = \mathbb{E}[v \mid \text{Sell Order}] \tag{2.9}$$

Since a buy order is more likely to originate from an informed trader who knows $v = V_H$, we have $a > \mathbb{E}[v]$. Conversely, a sell order suggests $v = V_L$, so $b < \mathbb{E}[v]$.

Adverse Selection and Spread Size

The resulting bid-ask spread $(a-b)$ is the MM's informational hedge. It is the required compensation to cover the expected loss incurred when trading with a better-informed counterparty. The half-spreads (δ_a, δ_b) increase with the proportion of informed traders (α) and the magnitude of the underlying informational uncertainty.

The model also demonstrates the dynamic nature of price discovery: a sequence of buy orders causes the MMs to continually update their belief that $v = V_H$ using **Bayes' Rule**. As this belief changes, the midprice $(\frac{a+b}{2})$ shifts, and the spread compensates the MM for the informational asymmetry.

2.3 Conclusion

The foundational models of market microstructure provide the essential economic theory for understanding the costs and dynamics of trading that drive Algorithmic and High-Frequency Trading strategy design:

- **The Grossman-Miller Model** identifies inventory risk as a primary source of the spread, leading to the cost of immediacy and the concept of **temporary price impact**.
- **The Kyle Model** establishes the mechanism of **price discovery**, quantifying the cost of **adverse selection** through **permanent price impact** (λ).
- **The Glosten-Milgrom Model** explains the necessity of the spread as an **informational hedge** that dynamically adjusts to the perceived toxicity of order flow.

These frameworks delineate the components of the trading cost, namely, Inventory Cost, Operational Cost (not explicitly modeled here, but necessary for competition), and Adverse Selection Cost—that any successful algorithmic strategy must manage. The following chapters will build upon these theoretical results to develop quantitative methods for optimal execution and market making.

Chapter 3

Models of Limit Order Books

This chapter is dedicated to a review of agent-based models of limit order books, that is, models depicting, at the individual agent level, possibly from a statistical point of view, the interactions that lead to transactions on a financial market. Far from being exhaustive, this survey is based on selected models that are representative of important, specific trends in agent-based modelling.

3.1 Early Order-driven Market Modelling: Market Microstructure and Policy

The earliest market simulations were designed to study regulatory issues. The first attempt, by Stigler [22], investigated the impact of SEC regulations on U.S. stock markets using historical data from the 1920s and 1950s. With the growing computerization of markets, Hakansson et al. [14] developed a simulator to test the feasibility of automated market making.

While we do not review the full microstructure literature (see O'Hara [20] and Hasbrouck [15]), we highlight a few foundational works that underpin modern order book modelling.

A Pioneer Order Book Model

Stigler's 1964 model is the earliest documented attempt to simulate a financial market. Motivated by criticisms of the SEC's 1963 Special Study [8, 9], Stigler proposes a simple simulation in which price evolves within a discrete grid of $L = 10$ ticks. Orders arrive randomly, namely, buy or sell with equal probability and uniformly over the price grid, and become market orders if they cross the best opposing quote. Orders not executed within $N = 25$ time steps are cancelled. All orders have unit size, and the maximum number of standing orders is N.

Slanina [21] later simulated a similar model with $L = 5000$ and $N = 5000$, showing that returns follow a heavy-tailed distribution with a power-law exponent of about 0.3, though limited by the artificial cutoff imposed by L.

Microstructure of the Double Auction

Garman [12] provides an early study of the continuous double auction that explicitly incorporates temporal order flow. Buy and sell orders arrive as independent Poisson processes with intensities λ and μ, respectively, and are of unit size. Orders crossing the best quote become market orders. The state of the book is represented as a vector of standing quantities at each price level. Although analytically tractable

in principle, the model quickly becomes complex, highlighting the need for simulation methods and numerical analysis.

The key question posed by Garman is structural: Does the double auction mechanism itself generate the leptokurtosis observed in empirical return series.

Zero-intelligence Models

The Stigler and Garman models belong to a broad class of "zero-intelligence" models, later formalized by Gode and Sunder [13]. In such models, orders are submitted randomly, without strategic behavior. Two variants are typically considered:

- *Unconstrained zero-intelligence traders*: random orders at random prices.
- *Constrained zero-intelligence traders*: random orders respecting a personal reference price, preventing trades at a loss.

Zero-intelligence markets exhibit surprising allocative efficiency but also unrealistic price volatility. Later critiques (e.g. [7]) argue that some of these results are artefacts of the imposed constraints.

These early models formed the basis for later econophysics-inspired order book models.

3.1.1 Order-driven Market Modelling in Econophysics

The Order Book as a Reaction–Diffusion Model

Bak, Paczuski and Shubik [2] proposed a reaction–diffusion model in which buy and sell orders are treated as particles diffusing on a discrete price grid. A transaction occurs when a buy and sell particle meet. While highly stylized, the model initiates the analogy between order books and interacting particle systems. The resulting price dynamics exhibit a Hurst exponent of $H = 1/4$, far from empirical values.

Introducing Market Orders

Maslov [18] extends the particle model by introducing fixed limit orders, market orders that remove the best quote, and cancellation mechanisms. Orders are placed around the prevailing price, and market orders generate price changes. This produces heavy-tailed returns, though price dynamics remain subdiffusive ($H = 1/4$).

Deposition–Evaporation Models

Challet and Stinchcombe [6] interpret the order book as an infinite one-dimensional lattice with deposition (limit orders), annihilation (market orders), and evaporation (cancellation). The volatility behaviour is closer to empirical findings than earlier models but still lacks over-diffusive properties.

3.1.2 Empirical Zero-intelligence Models

Mike and Farmer [19] represent a major advance in empirically calibrated zero-intelligence modelling. Orders are placed according to Student-t distributions centered around the best quotes. Cancellations depend on three empirically observed factors:

1. Position in the order book: $y(t) = \Delta(t)/\Delta(0)$;
2. Imbalance between bid and ask queues: $N_{\text{imb}}(t)$;
3. Total number of standing orders: $N(t)$.

The cancellation probability is given by:

$$P(C \mid y(t), N_{\text{imb}}(t), N(t)) = A(1 - e^{-y(t)})\big(N_{\text{imb}}(t) + B\big)\frac{1}{N(t)}, \quad (3.1)$$

where A and B are fitted to data.

The model reproduces several important stylized facts (heavy tails, realistic spread distribution), though it still encounters issues related to order-book stability and volatility clustering.

3.2 The Mathematical Structure of Zero-Intelligence Models

This section develops a general mathematical framework for modelling limit order books through Markovian, order-driven dynamics. Our objective is to understand how the structural features of the book, namely, its depth profile, order-flow composition, and quote adjustments, which govern the evolution of market observables such as price and spread.

We begin with a minimalistic agent-based representation of a Bachelier-type price process, which illustrates how classical diffusion models arise from microstructural order flows. We then introduce the full Markovian description of a limit order book driven by independent Poisson processes, including the evolution of queue sizes, best quotes, and price increments.

3.2.1 An Elementary Approximation: Perfect Market Making

To motivate the Markovian modelling of limit order books, consider a highly stylised "perfect market making" framework. Although simple, it reveals how macroscopic diffusion-type price behaviour emerges from discrete order-flow events.

We impose the following assumptions:

- The order book initially contains one unit-sized limit order at every price level above the best ask $P_A(0)$ and below the best bid $P_B(0)$. The initial spread is one tick.
- Buy and sell market orders arrive as independent Poisson processes $M^+(t)$ and $M^-(t)$ with respective intensities λ_+ and λ_-.
- A single liquidity provider maintains a one-tick spread. After a buy (respectively sell) market order, they submit a buy (respectively sell) limit order with probability u, and an order on the opposite side with probability $1 - u$.

The resulting mid-price process satisfies:

$$dP(t) = \Delta P\big(dM^+(t) - dM^-(t)\big)Z, \tag{3.2}$$

where Z is a Bernoulli random variable with:

$$\mathbb{P}(Z = 1) = u, \qquad \mathbb{P}(Z = 0) = 1 - u,$$

and ΔP denotes one tick.

The infinitesimal generator is:

$$Lf(P) = u\lambda_+\big(f(P + \Delta P) - f(P)\big) + u\lambda_-\big(f(P - \Delta P) - f(P)\big). \tag{3.3}$$

Expanding in Taylor series and grouping first-order and second-order terms yields

$$Lf(P) = \frac{u(\lambda_+ + \lambda_-)(\Delta P)^2}{2}\,\frac{\partial^2 f}{\partial P^2} + u(\lambda_+ - \lambda_-)\Delta P\,\frac{\partial f}{\partial P} + o((\Delta P)^2). \tag{3.4}$$

Under the scaling limits:

$$u(\lambda_+ + \lambda_-)(\Delta P)^2 \to \sigma^2, \qquad u(\lambda_+ - \lambda_-)\Delta P \to \mu, \tag{3.5}$$

the generator converges to the diffusion operator:

$$Lf(P) = \frac{\sigma^2}{2}\frac{\partial^2 f}{\partial P^2} + \mu\frac{\partial f}{\partial P}, \tag{3.6}$$

corresponding to a Wiener process with drift. This example illustrates how classical continuous-time models may arise as scaling limits of sim-

ple order-driven mechanisms.

3.2.2 Limit Order Book Representation and Notation

We now introduce the general stochastic framework for a limit order book driven by Poissonian order flows. The book is represented in a moving frame of fixed size K on each side of the spread.

Let

$$(a(t);\, b(t)) = (a_1(t), \ldots, a_K(t);\; b_1(t), \ldots, b_K(t)),$$

where

- $a_i(t)$ is the queue size i ticks above the best bid,
- $b_i(t)$ is the queue size i ticks below the best ask.

Queue sizes live in $q\mathbb{Z}$, where q is the lot size. Define cumulative depths:

$$A_i = \sum_{k=1}^{i} a_k, \qquad B_i = \sum_{k=1}^{i} |b_k|. \tag{3.7}$$

The indices of the first non-empty ask and bid queues are given by the generalized inverse:

$$A^{-1}(q') := \inf\{p : A_p > q'\}, \qquad B^{-1}(q') := \inf\{p : B_p > q'\}. \tag{3.8}$$

The spread in ticks is therefore

$$i_S = A^{-1}(0) = B^{-1}(0). \tag{3.9}$$

Boundary Conditions. To ensure that best quotes remain well defined and the state process remains Markovian, we impose constant boundary depths a_∞ and b_∞ at price levels leaving the moving frame. This prevents the book from emptying and keeps the spread bounded.

3.2.3 Order Arrivals and State Transitions

The evolution of the order book is driven by three types of events:

- Market orders, arriving as Poisson processes $M^{\pm}(t)$ with intensities $\lambda_M^{\pm}$;
- Limit orders at level i, arriving with intensities $\lambda_{L,i}^{\pm}$;
- Cancellations at level i, occurring with intensities $\lambda_{C,i}^{\pm} a_i$ and $\lambda_{C,i}^{\pm} |b_i|$.

All orders have fixed size q. Limit orders arrive relative to the best opposite quote on their respective sides.

The queue sizes satisfy stochastic difference equations combining:

- Increases from limit-order arrivals,
- Decreases from cancellations and market-order executions,
- *Shift operators* that reindex queues when a best quote is consumed or a limit order arrives inside the spread.

These shift operators maintain exactly K visible levels in the moving frame and ensure a consistent Markovian representation.

3.2.4 Price Dynamics

Let $P_A(t)$ and $P_B(t)$ denote the best ask and bid prices. Events at or within the spread affect these quotes:

- A market buy consumes depth on the ask side,
- A limit order placed inside the spread improves a quote,
- A cancellation at the best queue may widen the spread.

The best ask satisfies:

$$dP_A(t) = \Delta P\Big[\big(A^{-1}(q) - i_S\big)\, dM^+(t) - \sum_{i=1}^{K}(i_S - i)^+\, dL_i^+(t) \tag{3.10}$$

$$+ \big(A^{-1}(q) - i_S\big) dC_{i_S}^+(t)\Big]. \tag{3.11}$$

A symmetric expression governs $P_B(t)$. The mid-price $P(t) = (P_A(t) + P_B(t))/2$ and spread $S(t) = P_A(t) - P_B(t)$ follow:

$$\begin{aligned} dP(t) = \frac{\Delta P}{2}\Big[\big(A^{-1}(q) - i_S\big)\, dM^+(t) - \big(B^{-1}(q) - i_S\big)\, dM^-(t)\Big] \\ - \frac{\Delta P}{2}\sum_{i=1}^{K}(i_S - i)^+\, dL_i^+(t) + \frac{\Delta P}{2}\sum_{i=1}^{K}(i_S - i)^+\, dL_i^-(t) \\ + \frac{\Delta P}{2}\big(A^{-1}(q) - i_S\big)\, dC_{i_S}^+(t) - \frac{\Delta P}{2}\big(B^{-1}(q) - i_S\big)\, dC_{i_S}^-(t). \end{aligned} \tag{3.12}$$

Similarly, the spread dynamics satisfy

$$\begin{aligned} dS(t) = \Delta P\Big[\big(A^{-1}(q) - i_S\big)\, dM^+(t) + \big(B^{-1}(q) - i_S\big)\, dM^-(t)\Big] \\ - \Delta P\sum_{i=1}^{K}(i_S - i)^+\, dL_i^+(t) - \Delta P\sum_{i=1}^{K}(i_S - i)^+\, dL_i^-(t) \\ + \Delta P\big(A^{-1}(q) - i_S\big)\, dC_{i_S}^+(t) + \Delta P\big(B^{-1}(q) - i_S\big)\, dC_{i_S}^-(t). \end{aligned} \tag{3.13}$$

These expressions provide explicit links between the local liquidity profile of the book and the resulting dynamics of price and spread. Depth at the best levels determines the immediate impact of market orders, while the limit-order flow within the spread governs quote adjustments and spread behaviour.

3.3 Conclusion

These insights form the foundation for the next chapter, where we move from theory to practice by simulating limit order book dynamics under the modelling assumptions introduced here. Through numerical experimentation, we will investigate how the interaction of arrivals, cancellations, and liquidity profiles gives rise to the statistical properties observed in real markets.

Chapter 4

Numerical Simulations of Limit Order Book using Poisson Processes

4.1 Introduction

LOBs act as a mechanism for buyers and sellers to interact in financial markets. Today, most exchanges provide real-time data access. With the rise of algorithmic trading systems, the order book and its history have become one of the most utilized forms of financial data.

The Necessity of LOB Modelling

Modelling and simulating LOBs is necessary for several key reasons:

- **Understanding Market Dynamics:** The time evolution of the order book captures the highly complex dynamics of supply and

demand.

- **Observing Price Formation:** The LOB allows researchers to observe the price formation of a security at the most granular level possible.
- **Gauging Liquidity:** The state of the order book serves as a strong indicator of market liquidity.
- **Testing and Calibration:** Simulating the order book lets practitioners replay history. This is essential for backtesting, calibrating, and fine-tuning automated algorithmic trading strategies.

The recent availability of faster and cheaper computing power has allowed these simulations to grow richer. Models are generally tested against empirical observations, known as stylized facts, to ensure their outputs reflect real-world market behavior.

Methodologies in LOB Simulation

The current state of the art features a wide variety of simulation techniques. These models broadly fall into two main categories: Point Process models and Deep Learning models.

Point Process Models

Mathematically, an LOB is an aggregate of individual orders arriving at different times. Therefore, it is natural to model the order book as a queueing system using point processes. Because these models are mathematically descriptive, they are fully explainable.

- **Poisson Processes:** Poisson models assume that order arrivals are independent of each other. Basic versions are "zero-intelligence" models because they use no prior expert heuristics or market information. Researchers have enhanced these models by making order arrival rates variable, depending on the current state of the queue. While mathematically convenient, the assumption that all orders are independent contradicts real market behavior. For example, Poisson models cannot explain the clustering effect of order events.
- **Hawkes Processes:** Hawkes processes solve the major challenges of Poisson models by relaxing the assumption of independent increments. Instead, they model order flow as an endogenously excited process. This means past events can trigger future events. A Hawkes process can have multiple dimensions to represent different order types, and these dimensions can excite one another. Because of this, Hawkes models successfully reproduce important structural details like volatility clustering and implicit market impact. While highly adaptable, their complex math makes calibration a key challenge.

Deep Learning Based Models

The LOB contains several sources of complexity and non-linear dynamics. To handle this, researchers increasingly use large parameterized neural networks. While these models act as black boxes and lack explainability, they perform very well at reproducing stylized facts.

- **Predictive Networks (CNNs, RNNs, LSTMs):** Researchers

use structures like Convolutional Neural Networks and Long Short-Term Memory networks to predict mid-prices, future price directions, and order volumes. These models ingest historical tick data and are effective because they can capture complex, path-dependent feedback loops.

- **Generative Networks (GANs):** Generative Adversarial Networks and their variants are used to generate artificial financial time series. By conditioning the network on a window of historical orders or the current LOB state, these models can simulate highly realistic order streams and market transitions.
- **Large Language Models (LLMs):** Recent approaches treat LOB data similarly to text. Researchers tokenize order book messages and use autoregressive models to generate sequences of simulated order flow, much like an LLM generates a sentence.

While researchers have explored numerous other methodologies for simulating a Limit Order Book, this review will focus specifically on two prominent categories: Point Process models and Deep Learning-based models. By examining these, we provide a focused look at both the highly explainable, mathematical approaches and the complex, data-driven techniques used in modern LOB simulation.

In this chapter, we introduce the numerical procedures used to simulate the limit order book models developed earlier. Our objective is to translate the theoretical framework of zero-intelligence order flows into an implementable algorithm capable of generating realistic order book dynamics. We begin with the baseline simulator driven by independent Poisson processes for market orders, limit orders, and cancellations,

before extending the approach in later sections to more general settings.

4.2 An Algorithm for Poissonian Order Flows

We consider the zero-intelligence model and describe an event-driven simulation scheme that generates the order book path in event time. For simplicity, and in line with standard practice, we assume throughout this subsection that the order book is symmetric, meaning that the statistical properties of order arrivals are identical on the bid and ask sides. With this assumption, we omit the "±" superscripts and introduce the following quantities:

$$\lambda_L = (\lambda_{L,1}, \ldots, \lambda_{L,K}), \qquad \Lambda_L = \sum_{i=1}^{K} \lambda_{L,i},$$

$$\lambda_C(a) = (\lambda_{C,1} a_1, \ldots, \lambda_{C,K} a_K), \qquad \Lambda_C(a) = \sum_{i=1}^{K} \lambda_{C,i} a_i,$$

$$\lambda_C(b) = (\lambda_{C,1} |b_1|, \ldots, \lambda_{C,K} |b_K|), \qquad \Lambda_C(b) = \sum_{i=1}^{K} \lambda_{C,i} |b_i|.$$

The total intensity of all possible events given the current state (a, b) is then given by,

$$\Lambda(a, b) = 2(\lambda_M + \Lambda_L) + \Lambda_C(a) + \Lambda_C(b), \tag{4.1}$$

where the factor of two reflects symmetry between buy and sell market and limit orders.

Algorithm: Order book simulation with Poisson order flows

Require: Model parameters: K (number of visible limits), λ^M, $\{\lambda_i^L\}_{i=1}^K$, $\{\lambda_i^C\}_{i=1}^K$ (intensities of order flows), a_∞, b_∞ (hidden-book depths), random distributions $\mathcal{V}_L, \mathcal{V}_M, \mathcal{V}_C$ (volumes of limit, market, and cancel orders).

Simulation parameters: N (number of events), X_{init} (initial book state).

1: Initialization: $t \leftarrow 0$, $X(0) \leftarrow X_{\text{init}}$.

2: **for** $n = 1, \ldots, N$ **do**

3: Update cancellation rates:

$$\Lambda^C(b) = \sum_{i=1}^{K} \lambda_i^C |b_i|, \qquad \Lambda^C(a) = \sum_{i=1}^{K} \lambda_i^C a_i.$$

4: Draw waiting time $\tau \sim \text{Exp}(\Lambda(a,b))$, where

$$\Lambda(a,b) = 2(\lambda^M + \Lambda^L) + \Lambda^C(a) + \Lambda^C(b).$$

5: Draw event type according to the probability vector

$$\frac{(\lambda^M, \lambda^M, \Lambda^L, \Lambda^L, \Lambda^C(a), \Lambda^C(b))}{\Lambda(a,b)},$$

corresponding respectively to: buy market order, sell market order, buy limit order, sell limit order, cancellation of a sell order, cancellation of a buy order.

6: Draw event volume from $\mathcal{V}_M$, $\mathcal{V}_L$, or $\mathcal{V}_C$ depending on the event type.

7: **if** limit order **then**

Draw the price level i with probability

$$\frac{(\lambda_1^L, \ldots, \lambda_K^L)}{\Lambda^L}.$$

end if

if cancellation **then**

if ask cancellation **then**

Draw level i with probability $\dfrac{(\lambda_1^C a_1, \ldots, \lambda_K^C a_K)}{\Lambda^C(a)}$.

else

Draw level i with probability $\dfrac{(\lambda_1^C |b_1|, \ldots, \lambda_K^C |b_K|)}{\Lambda^C(b)}$.

end if

end if

$t \leftarrow t + \tau$; update the order book state according to the selected event.

Enforce boundary conditions:

$$a_i \leftarrow a_\infty,\ i \geq K+1, \qquad b_i \leftarrow b_\infty,\ i \geq K+1.$$

end for

The simulation routine mirrors the stochastic dynamics of the model, at each step, the type of the next event is sampled according to the relative intensities of all possible events, the waiting time is drawn from an exponential distribution with parameter $\Lambda(a, b)$, and the order book

is updated accordingly.

Performance of Simulation Studies

In this section, we assess the performance of the numerical limit order book simulator introduced earlier by comparing its outputs to empirical results reported in the literature. Our analysis focuses on the stock SCHN.PA (Schneider Electric), following the reference study, and we replicate the corresponding figures using parameters estimated directly from market data. This allows us to evaluate how well the zero-intelligence framework reproduces stylized microstructural features observed in real limit order books.

We begin by describing the estimation procedure for the model parameters, after which we examine simulation outputs and carry out a detailed ablation study contrasting the plots obtained in the thesis with those reported in the original paper.

4.3 Parameter Estimation

The parameters of the model are estimated from the dataset described in [1]. Throughout this section, we focus on results obtained for SCHN.PA during March 2011.

Let T denote the length of the daily observation window, and let N_T^M represent the total number of market orders (buys and sells) observed over this period. The empirical intensity of market orders is then esti-

mated as

$$\widehat{\lambda}^M = \frac{N_T^M}{2T}. \tag{4.2}$$

For limit orders, let $N_{i,T}^L$ denote the number of limit orders (both sides combined) submitted at distance i ticks from the best opposite quote during the interval T. The corresponding arrival intensity is estimated by,

$$\widehat{\lambda}_i^L = \frac{N_{i,T}^L}{2T}. \tag{4.3}$$

Cancellation intensities require normalization by the average depth at each level. If $N_{i,T}^C$ denotes the number of cancellations at distance i ticks over the same interval, and $\langle X_i \rangle$ denotes the temporal average number of shares resting at that level, then

$$\widehat{\lambda}_i^C = \frac{1}{2T} \frac{N_{i,T}^C}{\langle X_i \rangle}. \tag{4.4}$$

Daily estimates of λ^M, λ_i^L, and λ_i^C are then averaged across the 23 trading days of March 2011 to obtain the final parameter set used in the simulator.

Order sizes are modeled through log-normal distributions. For each order type, namely, market, limit, and cancellation, we compute the empirical distribution of observed volumes and fit a log-normal distribution via maximum likelihood. This yields parameter pairs (v_M, s_M), (v_L, s_L), and (v_C, s_C) corresponding to market, limit, and cancellation orders respectively.

Reference Parameters for SCHN.PA

For clarity, we reproduce here the parameter values reported for SCHN.PA [1]:

- Number of visible limits: $K = 30$
- Hidden liquidity parameters: $a_\infty = 250$, $b_\infty = 250$
- Log-normal volume parameters:

$$(\nu^M, s^M) = (4.00,\ 1.19), \qquad (\nu^L, s^L) = (4.47,\ 0.83), \qquad (\nu^C, s^C) = (4.48,\ 0.82)$$

- Market-order intensity:

$$\lambda^{M\pm} = 0.1237$$

- Limit-order intensities (first 10 levels):

$$\lambda_i^{L\pm} = \begin{pmatrix} 0.2842,\ 0.5255,\ 0.2971,\ 0.2307,\ 0.0826, \\ 0.0682,\ 0.0631,\ 0.0481,\ 0.0462,\ 0.0321,\ \ldots \end{pmatrix},$$

- Cancellation intensities (scaled by 10^3):

$$10^3 \lambda_i^{C\pm} = \begin{pmatrix} 0.8636,\ 0.4635,\ 0.1487,\ 0.1096,\ 0.0402, \\ 0.0341,\ 0.0311,\ 0.0237,\ 0.0233,\ 0.0178,\ \ldots \end{pmatrix},$$

- Average depth at level i (first 10 levels):

$$\langle X_i \rangle : \ \{276,\ 1129,\ 1896,\ 1924,\ 1951,\ 1966,\ 1873,\ 1786,\ 1752,\ 1691, \ldots\}$$

Comparison and Ablation Study

To evaluate the robustness of the model, we compare our reconstructed plots with those reported in the reference text.

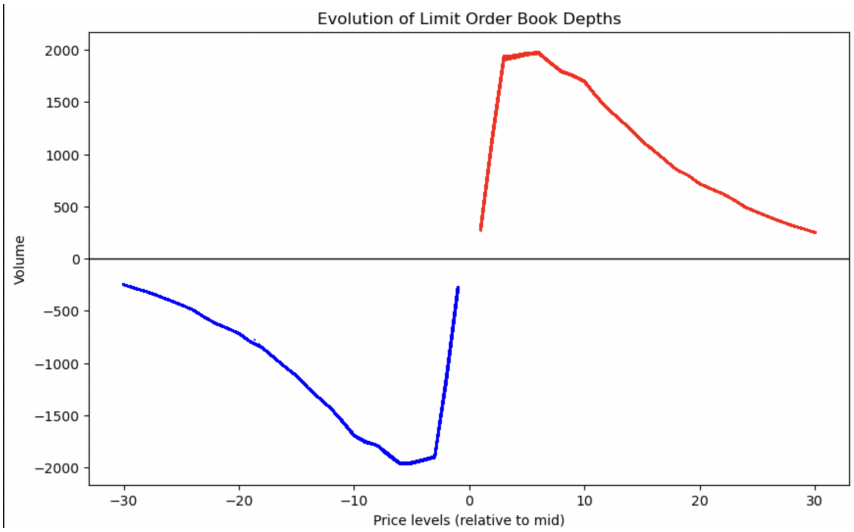


Figure 4.1: Simulated Average Depth of the Order Book for SCHN.PA w.r.t. the mentioned parameters

Further Simulations

The reconstruction of the depth profiles in the previous section demonstrates that the implemented simulator successfully reproduces the qualitative behaviour described in the reference zero-intelligence models. To further assess the robustness of the calibrated simulation framework, we extend our analysis to an entirely different dataset by studying the L10 millisecond-level limit order book of Microsoft (MSFT) on 21 June 2012, obtained from the LOBSTER database.

4.4 Data Description

The LOBSTER dataset provides message files and limit order book snapshots at user-specified depth. For a given trading day, the dataset includes:

- **Message File** (matrix of size $N \times 6$): containing the event time, event type (submission, cancellation, execution, etc.), order identifier, volume, price (in units of \$0.0001), and direction (buy/sell).
- **Order Book File** (matrix of size $N \times (4K)$): containing the evolution of the top K bid and ask levels, with associated prices and volumes.

Events are timestamped to millisecond precision (and up to nanoseconds for certain periods), and order book rows correspond exactly to the events listed in the message file, thus ensuring perfect synchronisation. Unoccupied book levels beyond the available depth are filled with dummy values following the LOBSTER standard (`-9999999999` for bids and `9999999999` for asks).

Using this structure, we first estimate the model parameters from MSFT data using the same methodology applied earlier for SCHN.PA, namely:

$$\lambda^M = \frac{N_T^M}{2T}, \qquad \lambda_i^L = \frac{N_{i,T}^L}{2T}, \qquad \lambda_i^C = \frac{N_{i,T}^C}{2T\langle X_i \rangle}.$$

4.5 Estimated Parameters

Applying the above estimators to the MSFT L10 dataset, we obtain the following Poissonian intensity parameters (buy and sell sides being treated symmetrically due to the large sample size):

$$\lambda^M_{\text{buy}} = 0.7139776, \qquad \lambda^M_{\text{sell}} = 0.7139776,$$

$$\lambda^L_i = \begin{pmatrix} 0.0,\ 2.8007,\ 2.2445,\ 0.6058,\ 0.3152,\ 0.2862,\ 0.2029,\ 0.1307, \\ 0.1258,\ 0.1872,\ 0.1216,\ 0.0213,\ 1.71 \times 10^{-4}, \\ 6.41 \times 10^{-5},\ 0.0,\ 6.41 \times 10^{-5},\ 0,\ \ldots, 0 \end{pmatrix},$$

and cancellation intensities:

$$\lambda^C_i = \begin{pmatrix} 0.0,\ 2.33 \times 10^{-4},\ 1.24 \times 10^{-4},\ 2.96 \times 10^{-5},\ 1.78 \times 10^{-5}, \\ 1.23 \times 10^{-5},\ 1.18 \times 10^{-5},\ 9.26 \times 10^{-6},\ 1.23 \times 10^{-5}, \\ 1.60 \times 10^{-5},\ 1.51 \times 10^{-5},\ 3.86 \times 10^{-6},\ 4.52 \times 10^{-5},\ 0, \ldots \end{pmatrix},$$

with larger cancellation spikes observed at deeper levels, reflecting the empirical behaviour of MSFT's order flow late in the trading day. Hidden liquidity parameters are set to

$$a_\infty = 250, \qquad b_\infty = 250,$$

consistent with the original zero-intelligence specifications.

4.6 Simulated versus Empirical Depth Profiles

Using the estimated intensity parameters together with the simulation algorithm described in Section 9.2, we generate synthetic order book depth profiles and compare them with the empirical MSFT depth computed directly from the LOBSTER dataset.

Despite originating from fundamentally different sources, one being an empirical millisecond-resolution order book and the other a stylised Poissonian zero-intelligence model, the two profiles show remarkable qualitative similarity. In both cases:

- Liquidity builds steeply near the best quotes,
- Depth peaks within a few ticks from the mid-price,
- A gradual decline is observed deeper into the book,
- Bid and ask curves remain approximately symmetric.

These results highlight the surprising effectiveness of the zero-intelligence framework in capturing the large-scale structural geometry of real-world order books, even when applied to a highly liquid NASDAQ stock with complex intraday microstructure dynamics.

Figures 4.2 and 4.3 present the empirical and simulated depth profiles respectively.

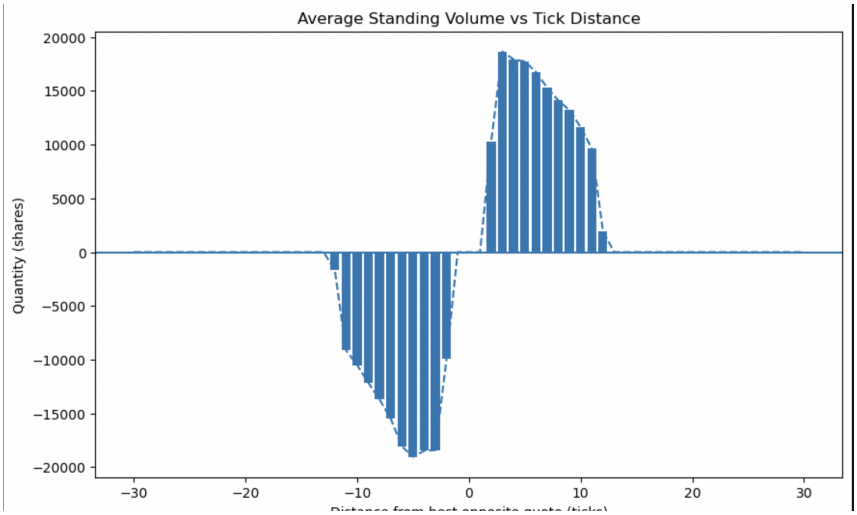


Figure 4.2: Empirical depth profile of MSFT (21 June 2012) computed from LOBSTER L10 data.

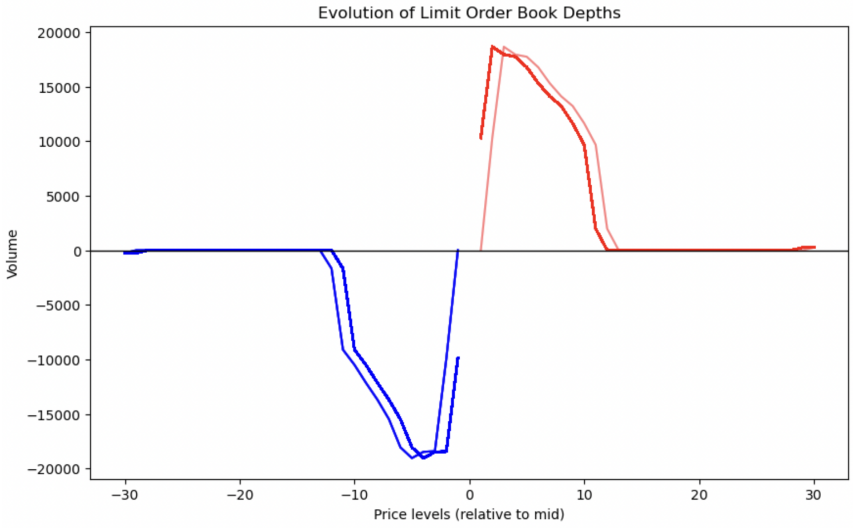


Figure 4.3: Simulated depth profile for MSFT using parameters estimated from the LOBSTER data.

The close alignment between the empirical and simulated structures strengthens the conclusion that the model, despite its simplicity, en-

codes sufficient microstructural features to replicate realistic order book behaviour across different assets and market conditions.

Chapter 5

Modelling Limit Order Books Using Hawkes Processes

In the previous chapters, we have seen LOB models dominated by Poisson processes, which assume that order arrivals are completely independent of one another. However, this assumption contradicts the reality of financial markets. Standard Poisson models fail to replicate critical microstructural phenomena, such as **volatility clustering** and **correlated order flow**.

- **Volatility clustering** in a limit order book refers to the tendency of large price changes to occur in bursts, followed by periods of relative calm. This arises because liquidity in the book is not constant. When the available depth at the best bid and ask is low, incoming orders have a larger price impact. During periods of heavy trading or uncertainty, limit orders are often cancelled and spreads widen, making the book thinner. As a result, successive trades move prices more, leading to sustained periods of high

volatility.

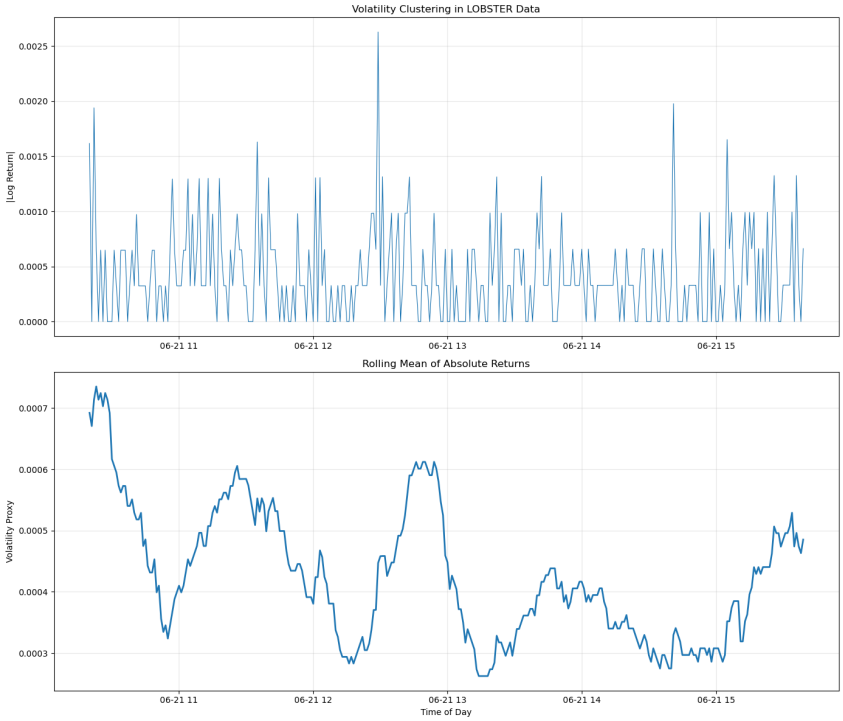


Figure 5.1: *Volatility Clustering shown by MSFT Stock on 21st Jun. 2012.* Figure depicts clear periods of high volatility followed by periods of low volatility

- **Correlated order flow** refers to the persistence in the direction of trades, where buy orders are likely to be followed by more buy orders, and similarly for sell orders. This occurs due to order splitting by large traders, algorithmic execution strategies, and common reactions to information. In a limit order book, such persistent order flow repeatedly consumes liquidity on one side, causing prices to move in that direction.

Here, the Hawkes process presents a compelling mathematical alternative. By dropping the strict requirement for independent increments, it

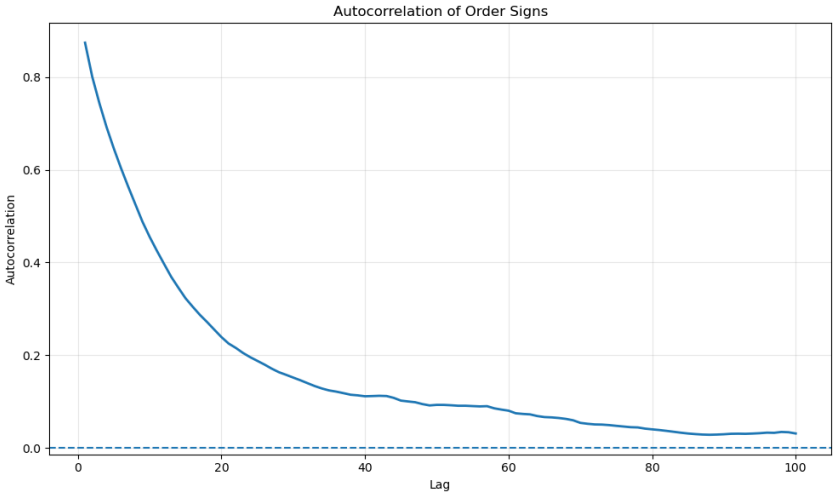


Figure 5.2: *Correlation between orders shown by MSFT Stock on 21st Jun. 2012.* For short lags, a high correlation is seen between orders

treats order flow as an endogenously excited process, meaning that past events can trigger future ones. This subtle shift allows the model to replicate both volatility clustering and correlated order flows. Hawkes processes also possess the advantage of being mathematically descriptive, as opposed to the 'black boxes' of deep learning models. They offer complete transparency, allowing researchers to see and explain the mechanics driving the simulation.

However, applying standard Hawkes models directly to LOB dynamics has its challenges.

- **Constant Order Sizes:** Standard Hawkes models focus primarily on capturing inter-event times, assuming order sizes remain constant. In reality, empirical distributions show significant spikes at round numbers (e.g., 100, 500), reflecting traders' preferences.
- **Negative Spreads:** The bid-ask spread in an exchange can never be mathematically less than one tick, a mechanical constraint that

standard point processes may violate.

- **Inhibitory Excitations:** Cross-excitations between certain events (like limit orders inside the spread and cancellations) can be inhibitory, potentially introducing mathematically invalid negative intensities into the point process.
- **Intraday Seasonality:** Order intensities exhibit a stationary "U"-shape throughout the trading day (high activity near open and close), which must be accounted for.

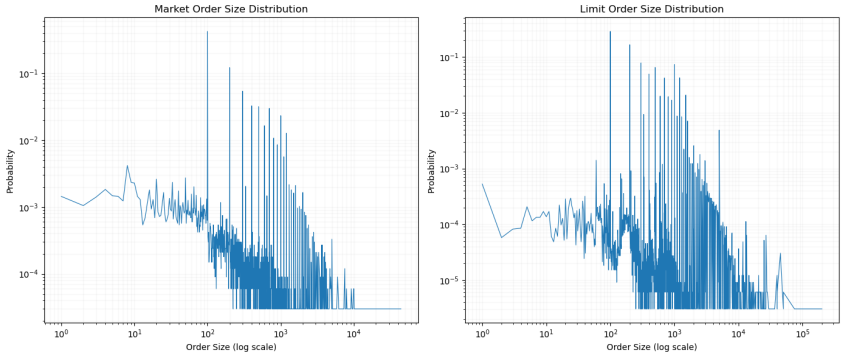


Figure 5.3: *(log-log) Empirical Distribution of Order Sizes of MSFT Stock on 21st Jun. 2012.* A high spike is seen on order sizes of round numbers such as 100/500/1000

The Compound Hawkes Process, augmented with specific mechanical constraints, effectively resolves these issues by sampling order sizes from a calibrated distribution and dynamically modifying the intensity functions.

5.1 Theoretical Framework

5.1.1 The Hawkes Process

For a d-dimensional Hawkes process, the intensity of the process $\lambda^{(i)}(t)$ and its associated counting process $N^{(i)}(t)$ for $i = 1, \ldots, d$ is defined as:

$$\lambda^{(i)}(t) = \mu^{(i)}(t) + \sum_{j=1}^{d} \sum_{t_j \in T_j} \phi^{(j \to i)}(t - t_j)$$

Alternatively, this can be formulated using integrals:

$$\lambda^{(i)}(t) = \mu^{(i)}(t) + \sum_{j=1}^{d} \int_0^t \phi^{(j \to i)}(t - u) dN^{(j)}(u)$$

Where:

- $T_j := \{t_j : t_j \leq t\}$ is the set of past event times in the j-th dimension.
- $\mu^{(i)}(t)$ represents the exogenous (baseline) intensity.
- $\phi^{(j \to i)}(t - t_j)$ represents the excitation kernel, a decaying function in time that dictates how an event in dimension j impacts the intensity of dimension i.

5.1.2 The Compound Hawkes Process (CHP)

A Compound Point Process (CPP) is mathematically defined as:

$$Z(t) = \sum_{i=1}^{N(t)} Y_i$$

Where $N(t)$ is a counting process with intensity $\lambda(t)$, and Y_i represents independent and identically distributed (i.i.d.) random variables denoting the jump size. A Compound Hawkes Process is a specific case of the CPP where the counting process $N(t)$ is driven by a Hawkes Process. In the context of LOB modelling, Y_i dictates the order size for a specific event type, independently sampled from a calibrated distribution parameterised by Θ.

5.2 Simulation Algorithm

To model the LOB dynamically, the researchers utilized a structured algorithmic approach focusing on the top levels of the book:

1. **Define the State Space:** Formulate the LOB as a 12-dimensional queuing system representing six key queues:

   $$\{ask_{+1}, ask_0, ask_{-1}, bid_{+1}, bid_0, bid_{-1}\}$$

   The 12 dimensions cover limit orders, market orders, and cancel orders at these respective levels. The subscripts $\pm$ denote the tick distance of the queue from the ask and bid price denoted by '0'.

2. **Enforce Mechanical Spread Constraints:** Prevent the spread from becoming negative by formulating the intensities of in-spread limit orders as a function of the current spread. Multiply the intensity by $(s-1)^{\beta}$, ensuring the intensity naturally drops to zero when the spread s is exactly 1 tick.

3. **Incorporate Time-of-Day Seasonality:** To capture the "U"-shape of daily trading volume, model the final intensity as a separable function. Multiply the Hawkes intensity by a piecewise constant function $f^{(i)}(Q_t)$, representing 30-minute bins throughout the trading day.

4. **Floor Inhibitory Kernels:** Apply a $\max(0, \dots)$ floor to the total excitation calculation to ensure that inhibitory cross-excitations do not push the point process intensities below zero.

5. **Non-Parametric Calibration:** Calibrate the Hawkes kernels non-parametrically using a linear-log spaced time grid, which captures the slow decay of the kernel functions over time.

6. **Ensure Process Stability:** Apply constraints during optimization so that the maximum eigenvalue of the kernel norm matrix is limited, ensuring the stationarity and stability of the simulated Hawkes Process.

7. **Parametric Fitting:** Fit the resulting non-parametric point estimates to a parametric function. Compare power-law kernels and exponential kernels using the Akaike Information Criterion (AIC), selecting the power-law function $\phi(t) = \alpha(1 + \gamma t)^{-\beta}$ for the final simulation.

8. **Calibrate Order Sizes:** Calibrate the probability distribution for order sizes using Maximum Likelihood Estimation. Use a geometric distribution combined with Dirac delta functions to insert artificial spikes at standard round numbers (e.g., 10, 50, 100).

9. **Simulation:** Utilize the thinning algorithm to generate synthetic order book events based on the finalized, calibrated parameters.

5.3 Key Experimental Results

As per the findings of Jain et al.[16], the methodology was tested against empirical equity data (such as Apple Inc. on the NASDAQ), yielding several notable results.

- **Replication of Stylized Facts:** The Compound Hawkes Process successfully captures complex microstructural facts, including the shape of the volatility Signature Plot, long auto-correlations of absolute returns, and the long-tailed nature of return distributions.
- **Accurate Spreads and Seasonality:** The explicit spread control mechanism accurately matches empirical spread distributions, while the time-of-day conditioning successfully replicates the "U"-shape activity seen in real order books.
- **Superiority Over Poisson Models:** When tested against a comparable 12D Compound Poisson model, the Poisson equivalent missed a large mass of inter-arrival time distributions, produced unrealistic spread distributions, and yielded an uncharacteristically flat volatility signature plot.
- **Concave Market Impact:** The simulator organically reproduces a concave relationship between Market Impact (MI) and total order quantity (Q_T), aligning with well-known empirical market observations. The simulator also correctly demonstrated that morning trading causes a higher market impact than afternoon trading.

Chapter 6

Simulating Limit Order Books with Generative Adversarial Networks

With the recent development in deep learning literature owing to the development of Large Language Models and architectures such as Convolution Neural Networks (CNNs), Long-Short Term Memory (LSTM), Recurrent Neural Networks (RNNs) and Generative Adversarial Networks (GANs), a lot of recent work features the use of such models to simulate the features of the Limit Order Book.

This chapter explores using Generative Adversarial Networks (GANs) to simulate these LOB snapshots. This chapter follows the architecture given by Cont et al. [10] and then conducts a simulation and ablation study on our own data. The goal is to learn the conditional probability distribution of a future LOB state given its current state.

6.1 Problem Setup

We represent the limit order book as a collection of queue sizes around the mid-price. This is called a centered LOB snapshot. In this representation, buy queues are recorded as negative numbers. Sell queues are recorded as positive numbers.

We model the transition of these queue sizes over a fixed time span. Price changes are naturally captured within this snapshot evolution. If the market price increases or decreases, the corresponding queues shift and change signs. A conditional model is absolutely necessary for this. The future state of the order book depends heavily on the current state.

6.2 The GAN Architecture

A Generative Adversarial Network consists of two competing neural networks. The Generator creates synthetic LOB snapshots from a random noise seed and a state condition. The Discriminator, or Critic, evaluates whether the generated snapshot is realistic compared to real market data. This model utilizes a Wasserstein GAN with Gradient Penalty (WGAN-GP). Gradient clipping is applied to the critic to stabilize the training process. The model operates in a Markovian setting. This means the condition fed into the network is simply the current LOB snapshot. Both the generator and the critic use multiple fully connected layers. LOB models must be symmetrical to avoid artificial arbitrage opportunities. We enforce symmetry during simulation. The model flips the LOB state with a fifty percent probability. This guarantees that unconditional price increases and decreases remain equally

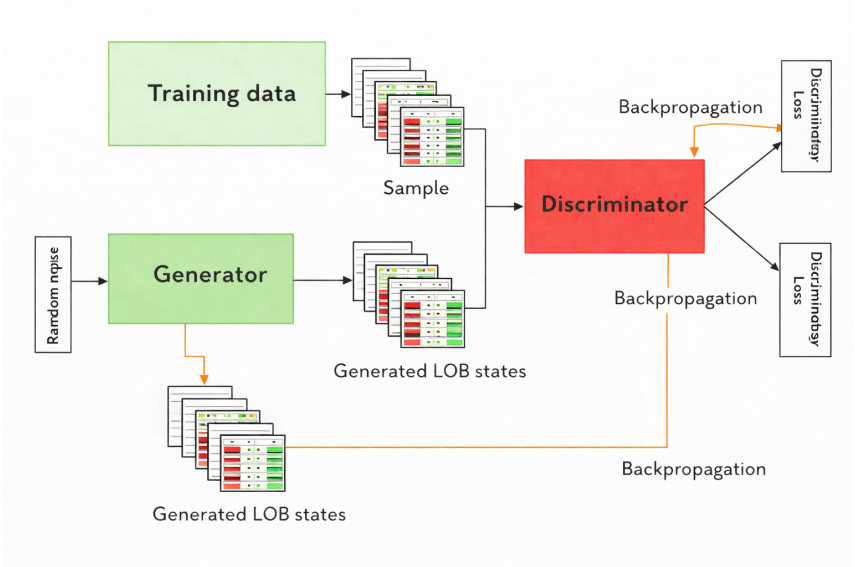


Figure 6.1: *Overview of the GAN Architecture*

likely, preventing structural price drifts.

6.3 Reported Results

Cont et al. [10] use L3 LOB data from 'lobsterdata.com'. They train their GAN on 20 days of MU Tick Data (MU is a large tick stock which rarely contains empty levels and mostly has a spread equal to the tick size) from 2016. They train their WGAN model for 300k iterations and establish some interesting results about some stylized facts.

6.3.1 Marginal Density

Results from Cont et al.[10] show that the marginal distributions of $X_{t+\Delta t}$, where the overall shape of the distributions is learned well by the model. For Ask-1 and Bid-1, the negative and positive values represent

Activation Hidden Layers	ReLU
Activation Final Layer	Linear
# of hidden layers	3
Hidden Layer Size	64
Δt	10 seconds
Days of data	1 day
# Samples	$\sim 40k$
Optimizer	Adam
Learning Rate	1×10^{-5}
Iterations	$\sim 20k$

Table 6.1: Table indicating attributes of the GAN architecture and training data used for training.

samples with price changes, where the queue is depleted and becomes a queue of the opposite sign. The generated samples show slightly less probability mass in these regions compared to the real samples, which indicates fewer generated price changes. The small bumps visible in Ask-2 and Bid-2 correspond to price changes of two ticks, which have a much lower probability mass. This shows that price changes of two ticks or more are much less likely to occur.

6.3.2 Average LOB Shape

Results also show that the average limit order book shape for the real and the generated data, and both match very closely. On closer observation, the average queue sizes are slightly larger for ask quantities.

6.4 Simulation and Ablation Studies

In order to further our understanding of the topic and test the performance of the model on external data, we simulate the model mentioned

in the paper on tick data of MSFT Stock (on 21st June 2012) obtained from LOBSTER. Figure 6.2 and 6.3 show the Marginal Densities obtained via our simulations. Our results are sub par as compared to the original paper, given the limited data and training resources, but they are still useful in establishing the validity of GANs in simulating LOBs and their properties.

6.5 Conclusion

Deep learning models are strong candidates for simulating limit order book dynamics because they can capture complex time evolution in Markov processes. Recent advances have also made training these models much easier. Many such models successfully reproduce important stylized facts and even show the concave market impact observed in real markets.

However, these models face major limitations. They are often black-box systems with low explainability, require careful hyperparameter tuning, and involve very high complexity with millions or billions of parameters. A more parsimonious and interpretable alternative is to model the transition of the order book state directly by taking the continuous-time limit and describing it through stochastic differential equations.

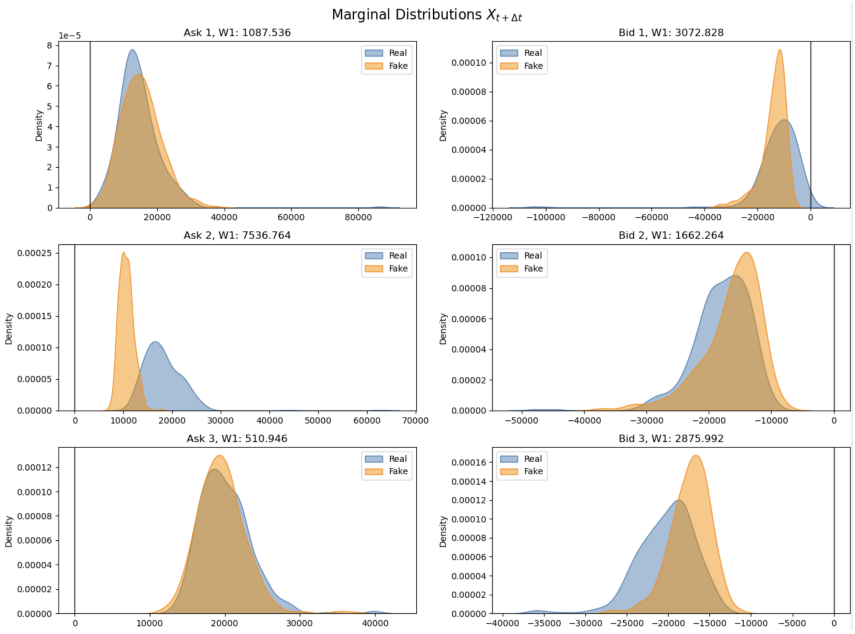


Figure 6.2: Marginal Densities of LOB Simulations based on MSFT Stock

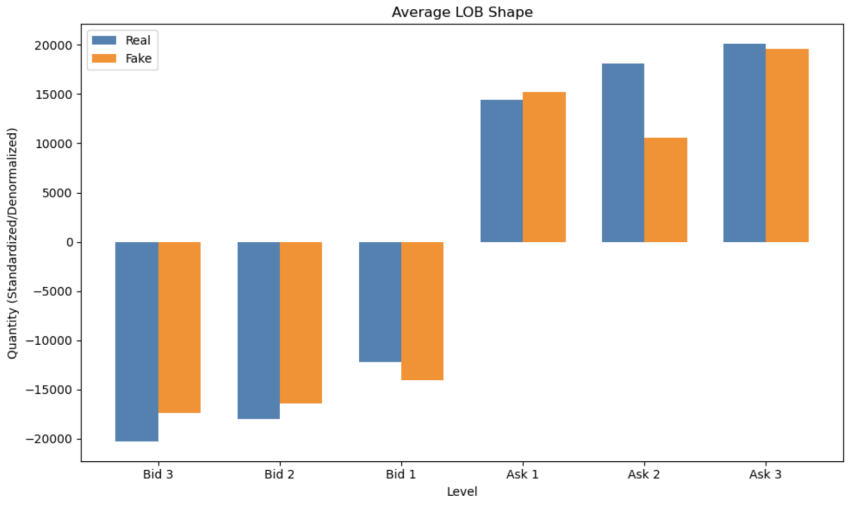


Figure 6.3: Average Shape of LOB Simulations based on MSFT Stock

Chapter 7

Mid-Price Prediction and Simulation Using Hawkes Processes

Following the analysis of Generative Adversarial Networks, we now explore point processes for mid-price prediction. This chapter models the limit order book using a calibrated Hawkes process. We simulate order arrivals and evolve the order book state to predict the mid-price path.

7.1 Problem Setup and Event Classification

We use historical message and order book data to calibrate our model. We extract a subset of 20,000 consecutive market events. The timestamps are normalized to start at zero.

We classify the order flow into a 6-dimensional system. This classification depends on the order type and trade direction. The six event streams are:

- Buy Market Orders (Executions)
- Sell Market Orders
- Buy Limit Orders (Submissions)
- Sell Limit Orders
- Cancel Bids
- Cancel Asks

7.2 Hawkes Process Calibration

We model the arrivals of these six events using a mutually exciting Hawkes process. We choose an exponential decay kernel with a fixed decay parameter ($\beta = 1.5$).

We estimate the model parameters using Maximum Likelihood Estimation. We calculate the log-likelihood over the empirical event streams. We minimize the negative log-likelihood using the L-BFGS-B optimization algorithm. This provides the baseline intensities (μ) and the 6x6 excitation matrix (α).

A key requirement is process stability. The Hawkes process is only stable if the spectral radius of the excitation matrix is strictly less than one. We enforce this constraint by calculating the eigenvalues and capping the maximum eigenvalue of the calibrated excitation matrix.

7.3 Order Book Simulation Algorithm

We simulate the arrival of future events using a standard thinning algorithm. The simulation runs up to a defined time horizon ($T = 100$).

We use these simulated events to evolve the state of the limit order book. We initialize the simulation using the real market bid price, ask price, and top-of-book queue sizes. The queues then update dynamically based on the simulated event types:

- Market orders decrease the opposite side queue.
- Limit orders increase the same side queue.
- Cancel orders decrease the same side queue.

A price change is triggered when a queue size drops to zero or below. The best price shifts by one tick. The depleted queue is then replenished. We determine the new queue size by randomly sampling from the empirical distribution of real market queues. Finally, we add a small amount of Gaussian noise to the calculated mid-price to reflect market micro-variations.

7.4 Simulation Results

We evaluate the simulator by comparing the generated synthetic data against the real market data.

7.4.1 Inter-arrival Times

We plot the distribution of time differences between consecutive events. The simulated inter-arrival times closely follow the real empirical distribution. The Hawkes process successfully captures the timing dynamics of the market.

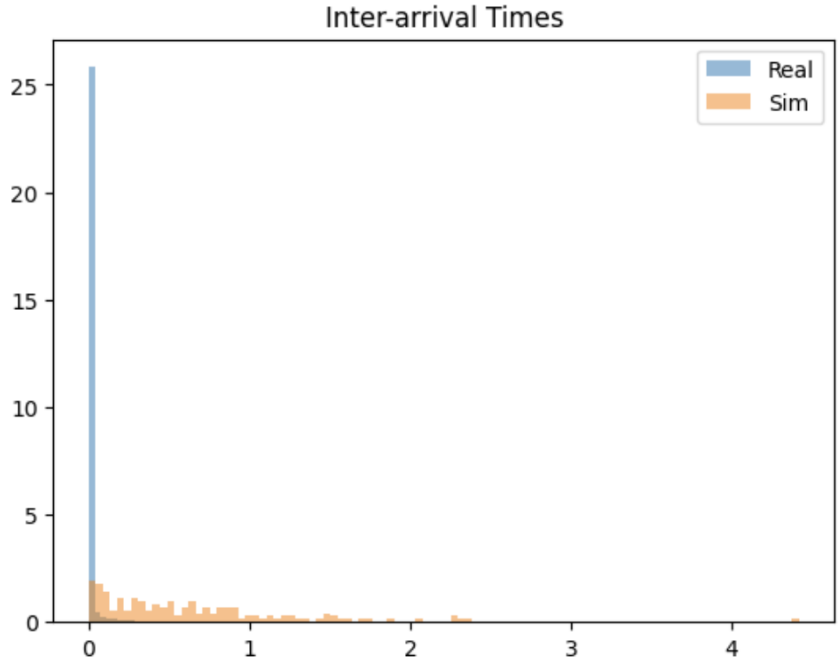


Figure 7.1: *Inter-arrival Times Distribution: Real vs. Simulated*

7.4.2 Return Distribution and Volatility

We calculate the price returns from the mid-price paths. The model struggles to perfectly match the empirical volatility. The real market volatility is calculated at roughly 0.0013. The simulated volatility is significantly higher at roughly 0.0125.

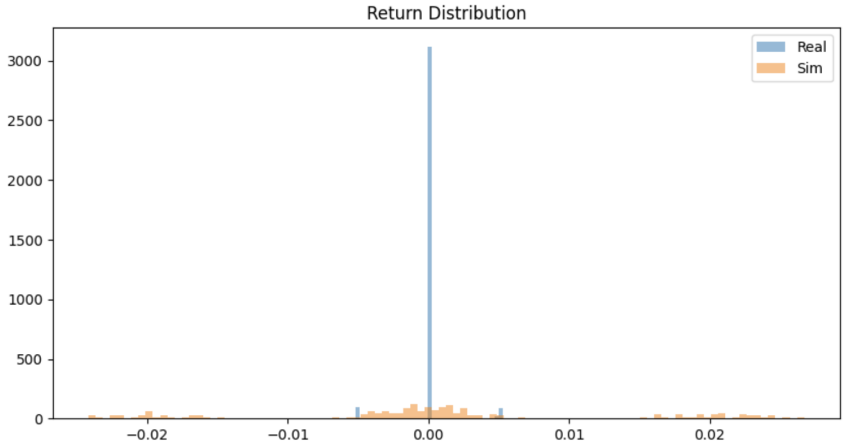


Figure 7.2: *Return Distribution: Real vs. Simulated*

7.4.3 Price Path Trajectory

The simulated price path captures the general microscopic movement but exhibits a wider overall range. The real mid-price ranges between 30.96 and 31.07. The simulated mid-price stretches from 30.93 to 31.15. The overestimation of volatility directly causes this expanded price range.

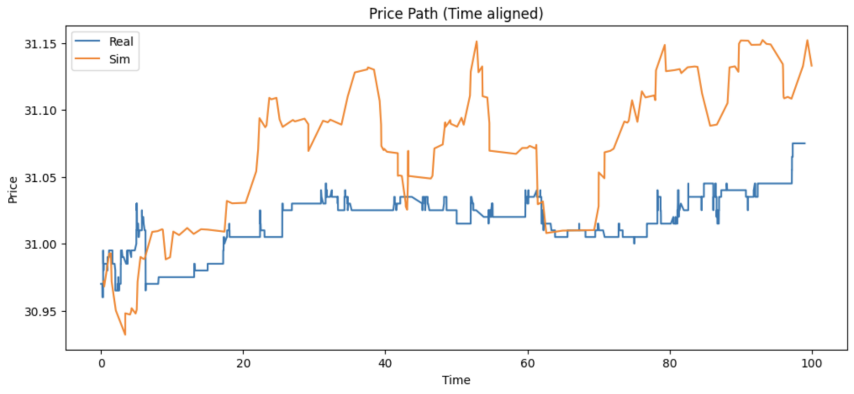


Figure 7.3: *Price Path Comparison over Time*

7.5 Conclusion

A 6-dimensional Hawkes process effectively models the self-exciting nature of order flow. It generates highly accurate event arrival times. However, this specific simulation framework overestimates mid-price volatility. Simulating realistic price paths requires more complex handling of order sizes and queue replenishment mechanics.

Chapter 8

Forecasting Limit Order Books with CNN-LSTM Architectures

As a future methodology to explore, they also mention a CNN-LSTM Architecture given by [11]. This work explores a combined CNN-LSTM network to classify future directions of the order book at high frequency, drawing heavy inspiration from the work of Zhang et al. [23] and Kolm et al. [17]. The primary goal is to predict whether the mid-price increases, decreases, or does not change in the very next observation.

8.1 Problem Setup

They represent the limit order book using the first ten levels of market depth. Each observation is a 40-variable vector containing the prices and volumes for the top ten bids and asks.

A sequence of raw limit order book states is a complex non-stationary process. To address this, they transform the raw data into order flow and order flow imbalance. Order flow is a 20-variable vector that computes changes between consecutive states while keeping the bid and ask sides separate. Order flow imbalance is a 10-variable vector calculated by simply subtracting the ask order flow from the bid order flow.

These formulas transform the sequence into a stationary process. This stationary property allows the LSTM to learn long-term dependencies much more easily. Furthermore, training on order flow gives the CNN layers the flexibility to combine bid and ask flows asymmetrically. Because of this, the forecasting models are expected to perform better on order flow than on order flow imbalance.

8.2 The CNN-LSTM Architecture

Different neural network architectures are highly complementary. When used together, they can learn unique structures in the data and improve the model's ability to execute a desired task.

The CNN layers first reduce the dimensionality of the multivariate input by aggregating the sides and levels of the order book. The output of these convolutions feeds directly into an Inception Module. This module wraps convolutions together to capture market behavior across multiple timescales. It serves as a moving average indicator where the decay weights are learned via backpropagation.

The outputs from the Inception Module are then concatenated and reshaped into a sequence. This sequence serves as the input to the

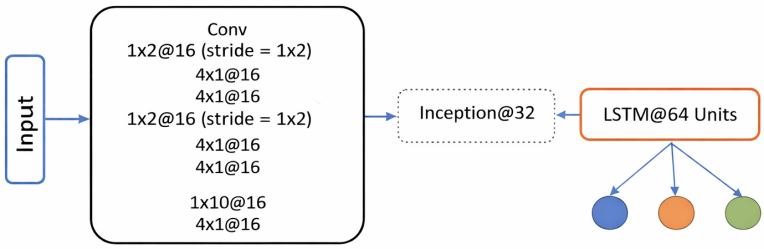


Figure 8.1: Architecture of the CNN LSTM Model as mentioned by Zhang et al. [23]

LSTM layer. The LSTM learns the temporal dependencies from this compressed parameter space, leading to a much more parsimonious model.

They formulate this forecasting problem as one of classification. They add an output layer with a softmax activation function to output the estimated probability for each price movement. The authors also apply Monte-Carlo (MC) dropout as a stochastic regularizer. This reduces overfitting and allows the model to take epistemic uncertainty into account.

8.3 Methodology and Training

The authors adopt a Box-Jenkins approach for modeling the time series. They initially fit Vector Autoregressive (VAR) models to the data.

They select the optimal lag order by minimizing the Akaike Information Criterion (AIC). This helps infer how long our LSTM lookback window should be.

They tune the dropout hyperparameters using a brute-force search looking for the best loss value from the dropout set of [0.1, 0.2, 0.3, 0.4, 0.5] for the Order Flow (OF) and Order Flow Imbalance (OFI) cases. During the training procedure, they apply early stopping to avoid overfitting. Training terminates automatically if the validation loss fails to improve for five consecutive epochs.

8.4 Trading Strategy Evaluation

To test the models, they perform 100 forward passes using MC dropout on the out-of-sample data, then average the predictions across these passes to approximate the true conditional probability of an outcome. They then compare two intra-second trading strategies. For each timestamp they trade at, they size our position at 30% of the volume available at the entry level.

The Softmax Strategy relies on a chosen probability threshold. The strategy goes long if the probability of an upward move exceeds the threshold, and it goes short if the probability of a downward move exceeds the threshold.

The Bayesian Strategy leverages predictive entropy to summarize classification uncertainty. Predictive entropy is minimized when the model is absolutely certain about one class. The entropy is maximized when probabilities are uniform across the classes. This strategy upsizes posi-

tions when the model is certain, downsizes positions when the model is uncertain, and exits the position entirely if the model becomes highly uncertain.

8.5 Results and Diagnostics

They compare the strategies using standardized profits and the Downward Deviation Ratio (DDR). DDR serves as our risk measure because it appropriately penalizes negative returns while rewarding positive returns.

Both models performed poorly in generating raw profits. The Bitcoin mid-price dropped by 1% over the duration of the trading period. Because the models were poor at predicting downward moves, they were doomed to behave poorly in this downward trending regime. Proper downsampling during training would have helped the model predict these downward moves.

Despite the trading losses, the Bayesian strategy remained much steadier than the Softmax strategy. This indicates that incorporating model uncertainty is highly beneficial. Normalized profits also clearly illustrated that training on order flow significantly outperforms training on order flow imbalance.

Finally, they performed model diagnostics using the Ljung-Box test. They tested the sample autocorrelations of the model residuals, computed via cross-entropy. The test provided statistical evidence to reject the null hypothesis of white noise. This indicates that the models underfit the training data. Financial data is incredibly complex, and it

is highly difficult for any interpretable model to perfectly fit the input data.

Chapter 9

Conclusion and Future Directions

Electronic exchanges and the Limit Order Book (LOB) form the backbone of modern financial markets. Modelling the dynamics of the LOB is a critical task for quantitative researchers and algorithmic traders. It allows practitioners to replay history, test strategies, and understand the fundamental mechanics of price formation.

This thesis investigated the core mechanisms of market microstructure and explored the methodologies used to simulate order book dynamics. We evaluated both mathematically descriptive models and modern, data-driven machine learning architectures.

9.1 Summary of Methodologies

Our investigation focused on two distinct modeling paradigms: Point Processes and Deep Learning architectures. Each approach presents

unique advantages and specific structural challenges.

9.1.1 Point Process Models

Historically, order arrivals were modeled using simple Poisson processes. However, these models operate under the strict assumption of independent increments. They fail to capture essential market realities like volatility clustering and correlated order flow.

To resolve this, we explored Hawkes processes. By relaxing the independence assumption, Hawkes models treat order flow as an endogenously excited process. Past events directly trigger future events. This framework successfully replicates the correlated nature of real market activity.

We specifically examined the Compound Hawkes Process. Standard point processes struggle with constant order sizes and the possibility of negative bid-ask spreads. The Compound Hawkes Process solves this by sampling order sizes from a calibrated empirical distribution. It also applies mechanical constraints to the intensity functions, guaranteeing that the market spread remains positive. While it accurately captured event arrival times, we noted that it can overestimate mid-price volatility.

9.1.2 Deep Learning Architectures

We then transitioned to deep learning. We explored Generative Adversarial Networks (GANs) to simulate order book transitions. By modeling the LOB as a centered snapshot, the GAN learned the condi-

tional probability distribution of future states. The Wasserstein GAN architecture successfully generated synthetic order books that matched the marginal densities and correlation structures of real market data. Our own ablation studies on MSFT tick data confirmed the validity of this approach, despite the constraints of limited training resources.

Next, we analyzed a combined CNN-LSTM architecture designed to forecast high-frequency mid-price directions. We transformed order book snapshots into stationary variables like order flow and order flow imbalance.

9.2 Explainability versus Complexity

A recurring theme is the fundamental trade-off between model explainability and predictive complexity.

Point process models are mathematically transparent. The parameters of a Hawkes process, such as the baseline intensity and the excitation kernel, have direct economic interpretations. Researchers can explicitly see how an aggressive market order impacts the arrival rate of future limit orders. However, these models require complex calibration techniques.

Conversely, deep learning models easily handle complexity and long-term dependencies. Architectures like the CNN-LSTM and the Conditional GAN excel at replicating empirical data distributions. However, they operate as black-box systems. They lack interpretability.

9.3 Future Avenues of Research

The literature suggests several promising directions for future research in LOB modeling.

The first is the integration of both paradigms explored in this thesis. Neural Hawkes processes replace fixed mathematical kernels with continuous-time LSTM units. This hybrid approach combines the transparency of a point process with the learning capabilities of a neural network.

Researchers are actively deriving Stochastic Differential Equations (SDEs) from the limits of complex Hawkes models. This translates discrete order book events into a macroscopic, continuous evolution of the order book state. It offers a more parsimonious alternative to neural networks.

Finally, the advent of Large Language Models (LLMs) provides a novel way to process LOB data. By tokenizing individual order book messages, researchers can train generative models to predict order flow sequences in the exact same manner an LLM predicts the next word in a sentence.

9.4 Concluding Remarks

Simulating the LOB is a highly complex but immensely rewarding computational challenge. There is no single perfect model. The choice of architecture depends entirely on the specific application, whether a researcher prioritizes the mathematical transparency of a Hawkes process or the high-dimensional predictive power of a deep neural network. As

computational power continues to grow and algorithmic markets evolve, the models used to simulate them will only become more realistic.

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